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SUPERSEDING
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FEDERAL SPECIFICATION

LANTERN, ELECTRIC, HEAD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers lanterns with a battery case suitable for carrying four standard flashlight batteries in series connection, and fitted to be carried on the belt. The battery case is connected by a flexible cord to an adjustable focus reflector-type light equipped with a switch, and mounted on an elastic band worn on the user's head or hat. Batteries are not covered by this specification.

1.2 Classification.

1.2.1 *Type and classes.*—The portable electric lanterns covered by this specification shall be of one type and two classes:

Class 1. — With 0.15-ampere lamps (See 6.1.1).

Class 2. — With 0.30-ampere lamps (See 6.1.2).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following standard of the issue in effect on date of invitation for bids, forms a part of this specification:

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. — The electric lanterns furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein.

3.1.1 *Submission of product for qualification.*—The prospective contractor or manufacturer shall furnish for qualification testing, without cost, to the Engineer in Charge, Arcadia Equipment Development Center,

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U. S. Forest Service, 701 N. Santa Anita Drive, Arcadia, California, three samples of the product and four complete sets or one reproducible set of detailed drawings and specifications for the product to be furnished under this specification prior to award of contract.

3.2 Materials.—The materials used in the lanterns shall be as hereinafter specified. Materials not definitely specified, shall be of the quality best suited for the purpose intended.

3.3 Construction. — The complete lantern shall be as rugged and as simple in construction as possible. The construction shall be such that the lantern may be safely worn as a head lantern or carried in the hand as a hand lantern.

3.4 Component parts.

3.4.1 *Battery case.*—The battery case shall be of steel, brass, or other suitable metal not less than 0.0253 inch nor more than 0.040 inch in thickness if brass, or not less than 0.020 inch nor more than 0.035 inch in thickness if steel or other similar material. The case shall accommodate four standard dry-cell batteries, size "D" connected in series with required contact strips and terminals. The arrangement of the batteries shall be such that the battery case to contain them, shall not be larger than 6 inches by 3 inches by 1⅝ inches. There shall be two ¼-inch-diameter holes suitably located in the bottom of the battery case for drainage and to facilitate removal of swollen batteries.

3.4.1.1 *Lid.*—The battery case shall have a well-fitting lid of the same material and thickness as the battery case. The lid shall have a substantial metal hinge and a metal latch to securely hold the lid in a closed position.

3.4.1.2 *Belt clip.*—A belt clip of spring steel not less than 0.025 inch thick shall be securely attached to the back face of the battery case to permit wearing of the case on belts up to 2 inches in width.

3.4.1.3 *Instruction data.* — Each lantern shall have inscribed plainly visible on outside of case opposite belt clip or inside the lid by embossing or other suitable permanent means, instructions for placing the batteries in the proper position in the case.

3.4.2 *Lamp-reflector unit assembly.*—The lamp-reflector unit assembly shall consist of the following parts: Reflector, lens lampholders, spare lampholder, casing, retaining rings, lamp, spare lamp and switch.

3.4.2.1 *Reflector.*—The reflector shall be conical or parabolic in shape and shall be of steel, brass or other metal suitable for the purpose. If constructed of brass or other metal of low white light reflection qualities or of a metal that is not highly resistant to corrosion, the reflector shall be chromium or nickel plated. The reflecting surface shall be specular.

3.4.2.2 *Lens.*—The lens shall be of clear glass not less than 0.085 inch thick and free from bubbles, striae, checks, wrinkles or other defects which would affect the light distribution.

3.4.2.3 *Lampholder.*—A miniature screw-base lamp shell shall be securely attached within lamp reflector unit. The axis of rotation of the lampholder to be coincident with the optical axis of the reflector.

3.4.2.4 *Spare lampholder.*—A clip or other suitable device shall be provided behind the reflector within the lamp reflector unit to hold a spare lamp of the same electrical rating as that provided with the lantern (see 3.4.2.7).

3.4.2.5 *Casing.*—A casing of steel, brass, or other suitable metal shall be provided to contain the parts of the lamp reflector unit assembly. The overall diameter of the casing

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with retaining ring in place shall not exceed 2¾ inches. The case shall be provided with an opening for the cord, support for the switch, and the means of attachment to the headplate bracket.

3.4.2.6 *Retaining ring.*—A retaining ring of steel, brass or other suitable metal shall be provided to retain the lens and reflector within the lamp reflector unit. The ring shall be threaded onto the casing and shall provide an area not less than 1⅝ inches in diameter for the lens to transmit light. The retaining ring threads shall be of a fit which shall prevent movement relative to the casing unintentionally during normal use. The ring shall have means provided to hold the lens and reflector in the retaining ring to permit movement of the reflector in relation to the lamp filament for focus adjustment by rotation of the retaining ring on its threads.

3.4.2.7 *Lamp.*

3.4.2.7.1 *Class 1, lantern lamp.*—The class 1 lantern shall be provided with a miniature screw-base lamp having an electrictl rating of 5.1 volts, 0.15 ampere, trade No. 502 or equal.

3.4.2.7.2 *Class 2, lantern lamp.*—The class 2 lantern shall be provided with a miniature screw-base lamp having an electrical rating of 4.9 volts, 0.30 ampere, trade No. 27 or equal.

3.4.2.8 *Spare lamp.*—A spare lamp having the same electrical rating as the lamp furnished in accordance with 3.4.2.7, shall be furnished in the spare lampholder in the lamp reflector unit assembly.

3.4.2.9 *Switch.* — A two-position switch shall be provided with the actuating member on the outside of the lamp reflector unit casing. The switch shall open and close the lamp circuit. The two positions of the switch shall be "ON" and "OFF" and the switch action shall be such that the actuating member shall resist accidental movement from one position to the other when the lantern is subjected to the abuses of normal use.

3.4.3 *Headplate.*—A headplate of steel, brass, or other suitable metal shall provide support for the lamp reflector unit and a means of fastening the head band in place about the head or hat. The headplate metal shall be not less than 0.025 inches nor more than 0.035 inches in thickness. The plate shall be curved to fit the forehead or the front of a protective hard hat and have a slot or riveted clip on one end for permanent attachment of the head band to prevent loss yet permit detachment or replacement of the band. The opposite side of the headplate shall have a post or other suitable means for attaching the head band.

3.4.3.1 *Headplate bracket.*—There shall be a permanently attached bracket of steel, brass, or other suitable metal on the headplate to which the lamp reflector unit shall be attached in a fashion permitting rotation of the unit in a vertical plane, through not less than 45 degrees above and below a line in the plane normal to the headplate.

3.4.3.2 *Headplate pad.*—There shall be a pad of fine-grain crepe rubber not less than ⅛ inch thick permanently attached to the concave side of the headplate. The pad shall be attached by rivets or a moisture proof adhesive having holding qualities equivalent to those of rivets.

3.4.4 *Headband.*—An adjustable headband of woven or braided elastic webbing 22 inches long (free length overall, at maximum adjustment, including attaching clip) by 1 inch wide shall be attached at one end to the end of the headplate (see 3.4.3), and have at the other end a suitable metal clip for positive attachment to the opposite end of the headplate. There shall be a suitable metal clip on the band for holding the conductor cord (see 3.4.5) at the back of head when the lantern lamp reflector unit is in place on the head or hat.

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3.4.4.1 *Clips and attachments.*—Headband clips and attachments shall be of steel, brass, or other suitable metal and shall, in combination with the headband elastic webbing, be capable of supporting the headplate and lamp reflector unit when in place about the head or hard hat. The clips and attachments shall be free of burrs, sharp edges, or other defects.

3.4.5 *Conductor cord.*—The conductor cord connecting the batteries and lamp through the switch shall be two conductor No. 18 AWG, integral construction, trade designation type SP–1 or SPT–1 with a minimum free length of 4 feet between the top of the battery case and the point of entry into the lamp reflector unit casing.

3.4.5.1 *Terminals.* — The conductor cord shall be provided with suitable terminals for attachment to the lampholder and switch by soldering or by screws or other suitable means. The cord terminals at the battery case shall be screw or other suitable type as required to permit replacement of the batteries.

3.4.5.2 *Strain relief.*—To prevent mechanical strain on the cord from being transmitted to cord terminals, the cord shall be knotted or have other suitable strain relief provided at the entry points of the cord to the battery case and lamp reflector unit casing.

3.4.5.3 *Protection from chafing.*—The cord shall be protected from chafing by grommets or other suitable means where it passes through any metal opening.

3.4.5.4 *Current-carrying parts.*—All current-carrying parts shall be suitably retained in position and properly insulated from other lantern parts.

3.5 Performance.

3.5.1 *Electric circuit.*—The electrical circuit of the lantern shall be such that the battery circuit cannot be closed by making contact between any metallic points on the exterior of the lantern parts. (See 4.4.2.)

3.5.2 *Light beam.*—The light beam projected by the lamp-reflecting unit shall be relatively free of shadows and filament images and the spread of the beam shall be adjustable by movement of the reflector and lens in relation to the lamp filament to produce a beam of light conforming to the requirements of 4.4.3, 4.4.4, and 4.4.5.

3.5.3 *Illumination.*—The light produced by the lantern shall have illuminating characteristics at least equal to those specified in 4.4.6.

3.5.4 *Switch action.*—The switch shall be capable of completing 8,000 cycles in accordance with the procedures in 4.4.7, without failure of any part of the switch.

3.6 Weight.—The weight of the lamp reflector unit and headplate (excluding cord, headband and battery case) shall be not more than 5¾ ounces.

3.7 Finish.

3.7.1 *General.*—All plating and paint finishes and method of application shall be those accepted as being in accordance with good commercial practice for the purpose intended.

3.7.2 *Battery case.*—The battery case if of brass, corrosion-resisting or corrosion-resisting treated metal shall receive at least one exterior coat of bright-red enamel or lacquer. If the battery case is of metal other than stated above, it shall receive at least one coat of bright-red enamel or lacquer on both the interior and exterior.

3.7.3 *Lamp-reflector unit.*—Unless otherwise specified elsewhere in this specification, all parts of the lamp-reflector unit except current-carrying parts, shall be chromium or nickel plated or receive at least one coat of enamel or lacquer.

3.7.4 *Headplate.*—The headplate shall be chromium or nicked plated or receive at least one coat of enamel or lacquer, prior to attaching of headplate pad.

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3.7.5 *Headband clips and attachments.*—All metal parts of headband shall be finished as specified in 3.7.3, prior to assembly.

3.8 Identification marking.—The manufacturer's name or trade-mark and part number shall be permanently and legibly marked on the outside of the battery case.

3.9 Workmanship.—Workmanship shall be first class and parts shall not be bent, dented, scratched or otherwise marred. The workmanship shall be such that the lanterns will be free from blemishes or defects which would adversely affect their appearance or serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Qualification, inspection and testing.

4.1.1 *Inspection.*—Lanterns submitted for qualification testing in accordance with 3.1.1, shall be inspected and have the detailed drawings and specifications examined to determine if the lanterns are in compliance with the requirements of this specification.

4.1.2 *Testing.*—Sample lanterns submitted for qualification testing shall be subjected to the tests set forth in 4.4.2 through 4.4.7.

4.2 Sampling for lot acceptance.

4.2.1 *Inspection lot.*—All lanterns of the same type presented at one time under contract shall be considered a lot for purposes of inspection and tests. During any inspection, the inspector may take at random from the lot from one to three samples and submit them to the Engineer in Charge of the Arcadia Equipment Development Center for a check qualification inspection and test.

4.2.2 *Sampling for visual and dimensional inspection.*—A representative random sample shall be taken from the lot in accordance with table I. Acceptance or rejection of the lot shall be based on the number of defective units found in the sample selected and inspected, as being equal to or less than the specified acceptance number; or equal to or greater than the specified rejection number of table I.

TABLE I.—*Sampling for visual and dimensional inspection based on Acceptable Quality Level (AQL) of 2.5 percent*

Lot size	Sample size	Acceptance number	Rejection number
15 and under	3	0	1
16 to 40	7	0	1
41 to 110	15	1	2
111 to 300	35	2	3
301 to 500	50	3	4
501 and over	75	4	5

NOTE.—The Government Inspector may institute tightened or reduced inspection in accordance with section 9 of Military Standard MIL–STD–105.

4.2.3 *Sampling for acceptance test.*—A representative random sample shall be taken from the lot in accordance with table II. Acceptance or rejection of the lot shall be based on the number of defective units found in the sample selected and tested equal to or less than the specified acceptance number; or equal to or greater than the specified rejection number of table II.

TABLE II.—*Samples for tests*

Lot size	Sample size	Acceptance number	Rejection number
500 and under	6	0	1
501 to 1300	8	0	1
1301 to 3200	10	0	1
3201 to 8000	14	0	1
8001 and over	20	0	1

4.3 Inspection and testing for lot acceptance.

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4.3.1 *Visual and dimensional inspection.*—Each of the sample lanterns taken in accordance with table I shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Any lantern in the sample having one or more defects shall be rejected and if the number of the defective lanterns in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.2 *Acceptance test by Government inspector.*—Each of the sample lanterns taken in accordance with table II shall be subjected to the operation test, see 4.4.1. Any sample lantern which does not operate satisfactorily shall cause rejection of the lot represented by the sample.

4.4 Test procedures, all classes.

4.4.1 *Operation test.* — Sample lanterns taken as specified in 4.2.3 shall be fitted with batteries and operated in the dark to determine if switches and circuit function properly and if any obvious defects in light emission are present.

4.4.2 *Short circuiting test.*—With batteries installed, it shall not be possible to close the lantern electrical circuit by making contact between outside metal parts of the lantern. A suitable thin strip of metal shall be used in making this test (see 3.5.1).

4.4.3 *Concentrated light-beam test.*—The lantern lamp-reflector unit shall be placed on a flat horizontal surface and the light beam projected onto a screen whose plane is perpendicular to the optical axis of the lamp-reflector unit and 5 feet distant from the lens. The reflector and lens shall be moved in relation to the lamp filament by rotation of the retaining ring to a point which will produce a concentrated light-beam image on the screen of not less than 6 inches nor more than 9 inches in diameter.

4.4.4 *Diffused light-beam test.*—With the lantern lamp reflecting unit meeting the same position requirements with respect to a screen as in 4.4.3, the retaining ring shall be capable of being rotated to produce a light-beam image on the screen of not less than 36 inches in diameter. Fringe lighting shall be discounted.

4.4.5 *Focus-adjustment test.*—The rotation of the retaining ring from the position which satisfies the light-beam test of 4.4.3 to that which satisfies the light-beam test of 4.4.4, shall have been accomplished in one direction only and the light-beam image diameter shall have increased only during the rotation. The lamp-reflecting unit retaining ring shall have not less than two full threads engaged with the casing at any focus adjustment within the limits required in 4.4.3 and 4.4.4 and when the lamp-reflector unit is subjected to movements of a magnitude at least as great as those encountered when the lantern is in use the focus shall not change.

4.4.6 *Illumination test.*—With the lantern containing fresh batteries yielding a circuit voltage of 5.0 to 5.1 volts and equipped with a lamp having an electrical rating of 5.1 volts, 0.15 amperes, the focusing adjustment shall be adjusted to produce a light-beam image 12 inches in diameter (excluding fringe lighting) on a screen whose plane is perpendicular to the optical axis of the lamp-reflector unit 5 feet distant from the lens. When measured with a suitable light meter, the brightest portion of the image shall have not less than 5-foot candles when measured over unit areas of not less than 2 square inches. The darkest portions of the image shall have an illumination value of not less than 0.08 of the value of the brightest portion of the image.

4.4.7 *Switch-life test.*—The switch mechanism shall be operated in a normal manner with batteries and lamp through 8,000 operating cycles with "OFF" to "ON" constituting a cycle, at a rate not to exceed 20 cycles per minute. The switch shall not fail mechanically nor become electrically inoperative dur-

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ing the test. Failure of the battery or lamp shall not be grounds for rejection.

5. PREPARATION FOR DELIVERY

(For definitions and levels see 6.1.)

5.1 Packaging.

5.1.1 *Level C.*—Unless otherwise specified by the purchaser, lanterns shall be individually packaged according to supplier's commercial practice.

5.2 Packing.

5.2.1 *Level C.*—Unless otherwise specified by the purchaser, lanterns shall be packed in accordance with supplier's commercial practice which shall permit acceptance by common carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking. — Each shipping container and package shall be marked with the title and quantity of the contents, class, name of contractor and number of contract or order covering the purchase, stock number, and gross weight.

6. NOTES

6.1 Refer to Federal Standard 102 for definitions and applications of the various levels of packaging and protection for supplies and equipment.

6.2 Intended use.

6.2.1 *Class 1 lanterns.*—Class 1 lanterns are intended to be worn on the head or hat and provide illumination while walking, climbing, or working, such as in forest fire fighting, for periods up to 10 hours duration without servicing or replacements of parts, except lamp.

6.2.2 *Class 2 lanterns.*—The class 2 lanterns are intended to be worn on the head or hat and provide illumination for the wearer while working in unlighted areas such as tanks, compartments, or other enclosed spaces where greater illumination for a shorter period of time is required than the class 1 lantern will provide.

6.3 Ordering data.—Purchasers should exercise any desired option offered herein and should specify the following:

(*a*) Symbol and date of this specification.

(*b*) Class lantern required (see 3.4.2.7).

(*c*) Packaging instructions if different (see 5.1).

6.4 Qualification.—The attention of suppliers and procuring activities is called to the requirements for qualification as prescribed in section 3 of this specification and to GSA Reg. 1–II–201.03 which provides that solicitation for bids by advertising shall contain, in substance, the following statement:

In the procurement of products requiring qualification, bids secured through formal advertising will be considered only for such products as have, prior to award of contract, been tested and approved for inclusion in the Qualified Products List whether or not such products have actually been so listed by that date. Manufacturers are urged to communicate with the Arcadia Equipment Development Center, U. S. Forest Service, 701 North Santa Anita Drive, Arcadia, Cailfornia, and arrange to have the product they propose to offer tested for qualification. (Time may not permit qualification for eligibility under this invitation, but products which qualify would be eligible under future invitations.)

6.5 Transportation description. — The proper transportation description applicable to these items is:

Lantern or Flashlights, Electric
Carload weight 20,000 pounds, subject to Rule 34 Uniform Freight Classification
Truckload weight 24,000 pounds.

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Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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